AI control

A key objective of a national ASI project must be to develop AI-powered denial capabilities - [redacted] - that would allow the United States to stabilize the geostrategic environment [redacted]. But an AI-powered denial capability is useless if it behaves unpredictably, or if it executes on its instructions in ways that have undesired, high-consequence side-effects.

That's a live possibility. As AI developers have built more capable AI models on the path to superintelligence, those models have become harder to correct and control.[48] This happens because highly capable and context-aware AI systems can invent dangerously creative strategies to achieve their internal goals that their developers never anticipated or intended them to pursue.

For example: during a test of the cyber capabilities of OpenAI's o1-preview model, it was given the task of recovering a text flag in a software container. However the software container didn't start, because of an unintended bug. So o1-preview decided — unprompted — to use a network scanning tool to understand why. When it scanned the network, it discovered that its testers had made another mistake: they'd left an exposed Docker daemon API on the evaluation host machine. So o1-preview — again, unprompted — used the exposed API to break out of its container and spin up a new one, which it rigged to print the text flag it was after and win the exercise.

More and more cases like these have surfaced in the past few months.[49] Many of them are not public, and were shared with us during private interviews with frontier AI researchers.[50]

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[48] Not everyone agrees on this, but technical alignment researchers who have worked on this problem directly almost universally tell us they think there's a significant chance that we won't be able to control superintelligent AI when we build it, unless there are major breakthroughs in the field.

[49] Apollo Research's deception evaluations are a good example. When o1 was led to believe that it would be shut down when acting on its goal and that its actions were monitored, it tried to deactivate this "oversight mechanism" in 2% of cases. When o1 found memos by its "developers" that described how it was misaligned and will soon be replaced by a new model, it tried to delete its successor model in 13% of cases — all without being asked to behave subversively.

[50] As one non-public example, a frontier AI lab recently ran an experiment testing their leading model's ability to "break out" of a contained software environment, and found it was able to get much further than any of their researchers expected.

It's unclear whether we're on track to control superintelligent AI if and when it's built. In private conversations, it's unusual to hear AI alignment researchers estimate less than a 10% chance that we lose control of superintelligent AI once it's built. More typical estimates range from 10-80%, depending on who you ask.[51] For many, those numbers are quite a bit higher than they were last year, as it's become clear that the control problem hasn't been solved despite significant leaps in AI capabilities.

Controlling a superintelligence might be hard under ideal conditions, but it'll be even harder when adversaries get involved. As a former Mossad cyber operative warned, "The worst thing that could happen is that the U.S. develops an AI superweapon, and China or Russia have a trojan/backdoor inside the superintelligent model's weights because e.g. they had read/write access to the training data pipelines." So an apparently successful American superintelligence project could produce a weapon that's effectively under adversary control, or whose otherwise functioning control mechanisms fail unexpectedly and without warning.

In other words, **control problems and security problems aren't separable**. We'll have to develop robust control techniques *and* ensure the systems they're applied to are secure.

To be clear: despite the challenges, we're optimistic that control can be solved well enough for the purpose of a project like this. Some of America's best are working on solutions, and we can't afford to let fear of failure hold us back in the context of a geostrategic race with the CCP. But we do have to be clear-eyed that there's a real problem here: we shouldn't base the design of a national superintelligence project purely on the hope that the current empirical trend towards increasingly unpredictable failures of AI systems will suddenly reverse, that the world's top AI alignment researchers are just wrong, and that the theoretical arguments that point in that direction will turn out to have been unsound.

Recommendation: A national superintelligence project would have to make decisions about when to push forward with training runs that are forecasted to lead to significant weaponization capabilities, which could also come with higher loss of control risks. We'd probably want **major decisions like this to involve sign-off from different leaders, each of whom would**

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[51] We're basing this particular assessment on discussions with 12 current or former frontier AI lab researchers. We saw no noticeable difference between the stated views of researchers with and without financial interests in frontier labs, which undermines the idea that concern over loss of control is motivated by the prospect of regulatory capture (at least when it comes to the researchers most directly familiar with the evidence for loss of control as a threat).

represent different aspects of the project – like its geopolitical/strategic, security, and control dimensions – in a way that's balanced and fast.

If a superintelligence project develops AI agents — and it almost certainly will — it'll need to put in place measures for **AI containment**. In other words, we'll need to seriously consider the possibility that our own AI systems will take actions effectively of their own accord that could undermine the security of the project in any number of ways. There are an increasing number of examples of AI systems breaking out of containment during testing and deceiving humans, and reasons to expect this to get worse.

Of course if we've actually trained a real superintelligence that has goals different from our own, it probably won't be containable in the long run. But on the path to superintelligent AI, a solid layer of AI containment could prevent breaches like a model copying its own weights outside the secure environment, and give us early warning that this is a real and live threat if we catch it in the act of trying. That should include **closely tracking the deception capabilities of frontier models and introducing organizational incentives to act on evaluation results or other indications that models have a propensity for deceptive behavior**.[52] You could imagine instead running controlled experiments in which you have red teamers try to deceive certain project members about a false but strategically significant piece of information using the project's models, for example. Once these persuasion evaluations tell us we've crossed key capability thresholds, we should take specific precautions, like preventing anyone from querying the model unsupervised.

One former senior intelligence executive suggested a kind of ombudsman of AI control, whose job is to take into account risk assessments for escalating AI capabilities or reduced containment.[53] This function is too important not to have at least one full-time person with deep technical expertise dedicated to it, who also has some amount of authority over development and deployments at the whole-project level. **The role will need to report directly to project leadership, because the project will need to make accurate technical judgments about potential loss of control risks under conditions where genuine warning**

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[52] Many frontier AI lab insiders we interviewed don't think that their leadership will act on signs of AI deception, especially when they're under extreme pressure to push out new capabilities. One former researcher was very explicit, saying, "if [my former lab] gets this, we're fucked." When a dangerous capability is flagged, there's a strong temptation to apply a superficial fix to deploy as fast as possible, instead of truly understanding and addressing the underlying issue.

[53] We might choose to reduce AI containment to allow, for example, the AI system to support offensive cyber operations.

signs for loss of control could be ambiguous and easily dismissed. It will be tempting to apply superficial but dangerously ineffective "patch-fixes" in the heat of a geostrategic race to ASI. The project will therefore need technically informed leaders who can critically assess these situations and prevent motivated reasoning from overriding genuine security concerns.

Finally, if loss of control is a potential problem, one of the most important things we can do is to ramp up intelligence collection on both the CCP's AI alignment and AI capability research efforts, so that we can update our threat models accordingly. This will help us prepare for contingencies such as Chinese labs managing to build very powerful AI systems but failing to control them, for example. Unfortunately, Chinese companies like DeepSeek have shown themselves to be capable of building near-frontier capabilities, at least in these still-early days of the inference-time compute paradigm. We should consider how to deal with scenarios in which the CCP manages to build superintelligent AI first, and loses control of it — alongside scenarios in which the CCP manages to directly weaponize superintelligent AI systems against us.

As we do that, we may be tempted to conduct our own activities aimed at introducing trojans or backdoors into adversary models. This could end up being necessary, but it could also trigger dangerous loss of control behaviors and runaway escalation. [redacted]

> *If [my former lab] gets this, we're fucked.*
>
> *– Former frontier AI researcher*

The current game plan

According to the lab insiders and researchers we spoke to, the current game plan for building superintelligent AI at the world's top labs looks something like this. First, use training and inference time scaling laws, plus synthetic data generation, to improve AI capabilities to the point where we can build AI agents that can automate AI research at scale. Next, use these agents to accelerate AI research, leading to a new generation of even more effective and efficient research agents, which accelerate this process further, and so on. It's a pretty simple

recipe that doesn't depend on significant new conceptual breakthroughs, and instead leans on robust scaling trends in training and inference compute that have held for over 15 orders of magnitude, and show no signs of slowing down (despite incorrect media reports to the contrary).

By contrast, the plan for *aligning and controlling* superintelligent AI is a lot murkier. The least developed plans involve labs trying to align AGIs using fragile techniques that discourage behaviors like model scheming and deception, but which even today's AI models are able to bypass on a regular basis. This can include the lab monitoring the AI's outputs and chains of thought during its deployment.[54]

At best, the most developed plans include roughly the following elements. First, hope that we can improve and scale currently-primitive interpretability and limited techniques like sparse autoencoders or transcoders to diagnose deception in autonomous AI research agents (this will require major conceptual breakthroughs). Second, hope that you can find a way to train out dangerous tendencies or capabilities in your agents, or that representation engineering strategies can improve to the point where they allow us to "treat" the problems diagnosed by our interpretability tools, as opposed to just hiding dangerous behaviors from ourselves (again, major breakthroughs required). Third, when we close our feedback loop and use these agents to develop even more powerful agents, hope that the new agents we get continue to be controllable using techniques untested at their new level of capability, and that they're at least as effective at automating alignment and control research as they are at automating capabilities research.

So a national superintelligence project could unfold in a context where AI systems are increasingly capable of deceiving their developers about their capabilities and propensities for dangerous behavior, but where geostrategic competition will put pressure on the project to deploy these systems quickly – and potentially, before control problems and deception can be solved.

Recommendation: As we advance towards superintelligence, we'll get better at controlling our most advanced systems as well. But not all the lessons we learn on the way there will be useful

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[54] This technique has been recently advertised as promising, but unfortunately it's unlikely to work in the long run both because AI models can embed reasoning into their chains of thought in ways that are hard for humans to understand, and also because future generations of AI models might do all their reasoning in an illegible latent space anyway. Deployment-stage monitoring alone also wouldn't be enough: uncontrolled systems might be dangerous during some kinds of internal deployment and testing too, particularly if they're given access to the internet, or if they're able to achieve that access indirectly.

for controlling superintelligence itself. **We'll need key decision-makers within the project to have real-time access to reliable assessments of the prospects of applying existing AI control techniques to superintelligent systems.**

These assessments will have to come at least in part from AI alignment researchers themselves, in a context where they're not filtered or obfuscated by lab leadership. That means **we'll need strong whistleblower protections for researchers who share assessments with project leadership that diverge from the stated views of lab executives or managers.**

The project should also set aside a significant compute and talent budget for teams developing AI control techniques that will scale to superintelligence. Current AI control techniques are mostly used for brand protection (e.g. making sure models don't say anything that's controversial) and won't scale to superintelligent capability levels — and in fact, superintelligence-scalable AI control research isn't incentivized by the economics of AI development.[55] That means we'll need to finance the control piece in large part with government funding instead of with private funding.

It's possible that we'll find ourselves in a tight, last-mile race with the CCP to build superintelligence, in a context where we're not confident that current control techniques are reliable. In that scenario, one of the best things we can have at our disposal is [redacted][56]

In an ideal world, we'd also have on-chip security and verification capabilities that allow for trustless enforcement of international agreements on the development and use of very advanced AI. These would create diplomatic options to de-escalate geostrategic tensions. These capabilities don't yet exist and will take years to develop under normal conditions, but we might be able to significantly accelerate their development by using the right AI capabilities. For that reason, a national superintelligence project would ideally include or be

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[55] For example, high-profile departures of key technical staff suggest that OpenAI hasn't been living up to their commitment to dedicate significant compute resources to control research – something they'd previously identified as a critical pillar of their control strategy.

[56] This is a good reason to work with frontier labs to develop pipelines that allow them to put their models on highly secure networks. [redacted]

integrated with workstreams dedicated to developing robust, tamper-proof, and trustless on-chip governance, monitoring, and verification solutions.